

Mathilde Perez

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Education

Télécom Paris

PhD in Graph Learning, Fairness in AI

- Fairness constraints on evolutionary graph learning models
- Evaluation framework for biased vs. unbiased link prediction models

*Palaiseau, France
Sept 2023 – present*

Université Paul Sabatier

Master in Applied Mathematics for Engineering, Industry and Innovation

- AI, Deep Learning, Machine Learning, Big Data, image processing
- Optimization, Statistics, Stochastic Simulations, PDEs, Signal Processing

*Toulouse, France
2020 – 2022*

Université Paul Sabatier

Bachelor in Applied Mathematics

*Toulouse, France
2017 – 2020*

Lycée des Arènes

Baccalauréat in Sciences

- Honors (Mention Bien), European English section

*Toulouse, France
2016 – 2016*

Publications

How Predicted Links Influence Network Evolution: Disentangling Choice and Algorithmic Feedback in Dynamic Graphs

2026

Mathilde Perez, Raphaël Romero, Jefrey Lijffijt, Charlotte Laclau

arxiv.org/abs/2603.03945 (UAI 2026 · Spotlight presentation, ECAF 2026)

TopoFair: Linking Topological Bias to Fairness in Link Prediction Benchmarks

2026

Lilian Marey, Mathilde Perez, Tiphaine Viard, Charlotte Laclau

arxiv.org/abs/2602.11802 (Preprint)

SimHawNet: a modified Hawkes process for temporal network simulation

2025

Mathilde Perez, Raphaël Romero, Bo Kang, Tijl De Bie, Jefrey Lijffijt, Charlotte Laclau

[10.1007/s10618-025-01119-1](https://doi.org/10.1007/s10618-025-01119-1) (Data Mining and Knowledge Discovery)

Experience

Research Engineer

Télécom Paris

- Simulation of temporal graphs with Hawkes processes

*Palaiseau, France
Apr 2023 – Sept 2023*

Human-Computer Interaction Intern

ISIR — Sorbonne / CNRS

- Modeling social attention through gaze in a conversational agent (Greta platform)

*Paris, France
Apr 2022 – Oct 2022*

NLP Intern

CerCo — CNRS

- Similarities between CLIP and the brain

*Purpan, France
May 2021 – July 2021*

Oral Presentations

- **SimHawNet: a modified Hawkes process for temporal network simulation** — ECML-PKDD 2025, Porto, Portugal · [Slides](#) & CAp 2024, Lille, France

- **How social media algorithms reinforce inequality** — [EUNIC Science Day](#), Madrid, Spain · [Slides](#)

Skills

Programming

Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, C/C++/C#, R, LaTeX, Unity, Java

Machine Learning & AI

Graph Learning, Deep Learning, Fairness in AI, Temporal Graphs, Hawkes processes

Soft Skills

Teamwork, Public speaking, Autonomy

Languages

French

Native

English

Advanced